

BRIGHTLEARN DATA ANALYTICS

Research Assignment 1

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Course: Data Analytics (September Intake)

SECTION A: DATA FUNDAMENTALS

(i) Database - an organised collection of data that is stored electronically so it can be easily accessed, managed and updated.

(ii) Why databases are essential?

Without databases the modern world could not function. Every time a person swipes a card, books a flight or visits a doctor, a database is working in the background.

(iii) Two real-world examples of where databases are used are:

(a) At Eskom: all permit to work records, maintenance logs and compliance data were stored in a database so engineers could retrieve any record instantly

(b) At a bank: every customer's account balance, transaction history and personal details are stored in a database updated in real time.

	Name	Date	Amount
Row 1	Mollie	01-09-26	R3000
Row 2	Agnes	05-09-26	R2900
Row 3	Jan	08-09-26	R3500
Row 4	Phillip	10-09-26	R2550

Database

2. Data Warehouse

This is a large centralised storage system holding historical data collected from multiple databases. Unlike a regular database which handles daily transactions, a Data Warehouse is designed specifically for analysis and reporting.

(iii) Company would use a data warehouse because:

- It allows them to

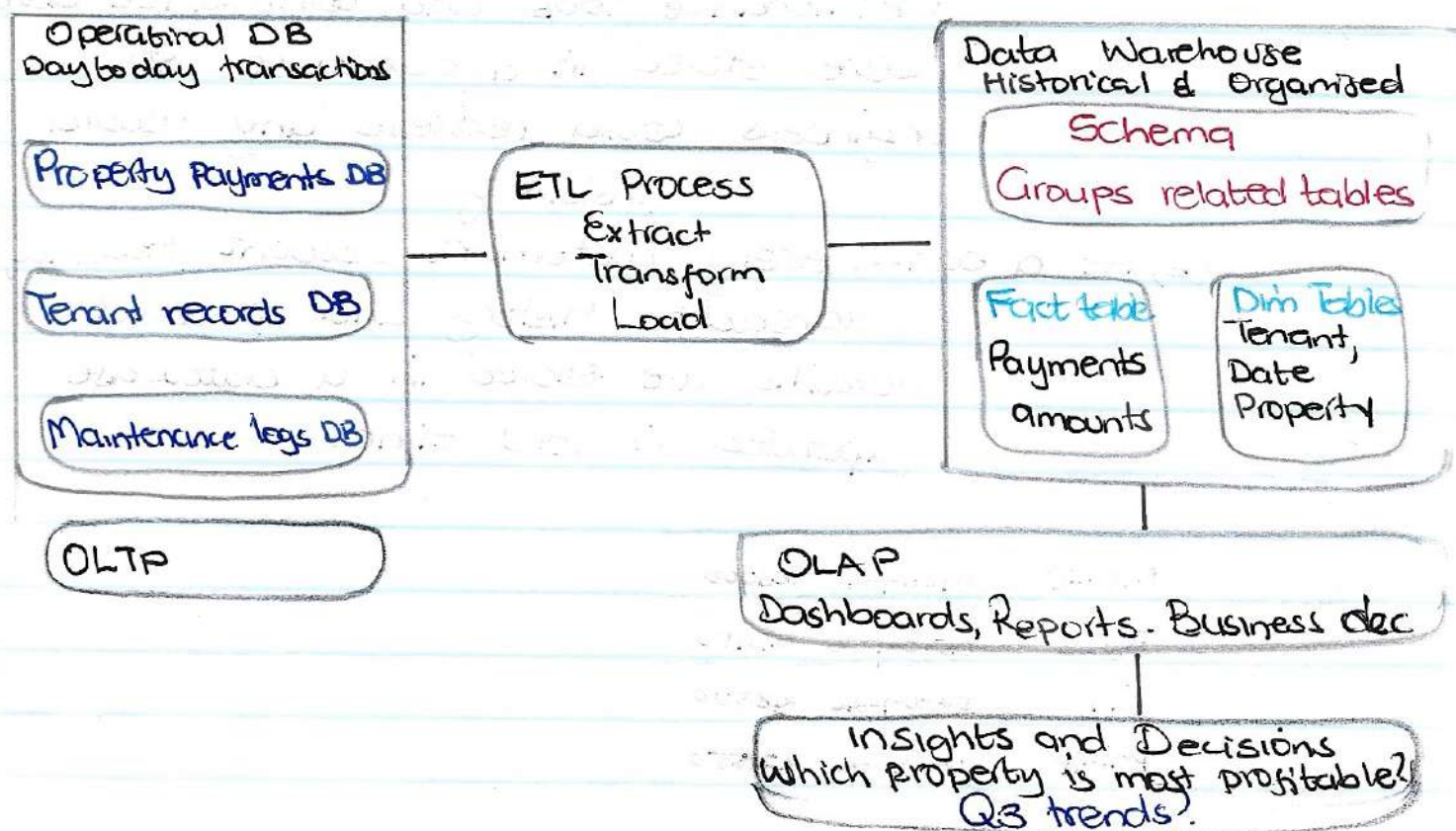
a) Analyse historical trends which can assist in seeing how things have changed over months and years.

b) Make better decisions

c) Combine data from many sources - instead of checking five different databases

Separately, everything is in one place.

d) Run reports without slowing down daily operations

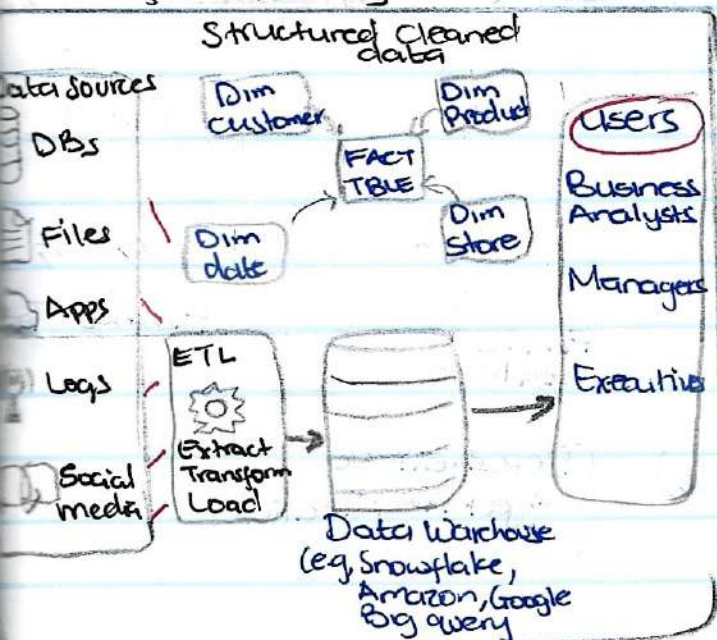


3. Data Lake

- (i) It is a large storage system that holds large amounts of raw data in many formats. The raw data can be structured, semi-structured and unstructured.
- (ii) Unlike a Data Warehouse which stores cleaned and organised data, Data Lake stores everything as it is, structured, semistructured and unstructured data all together.
- (iii) A business would prefer a data lake over a data warehouse when it collects a huge volumes of different data types like social media posts, videos, emails, sensor readings and want to store everything first and decide how to use it at a later stage.

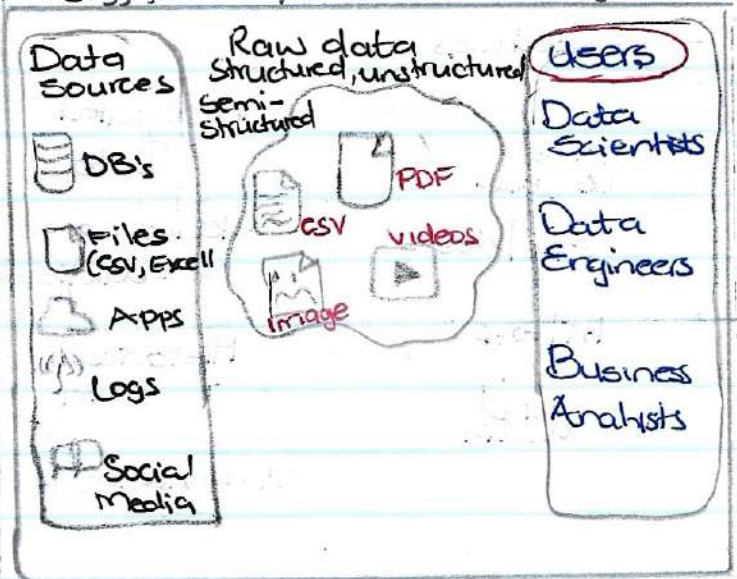
Data Warehouse

Stores cleaned, structured data for reporting



Data Lake

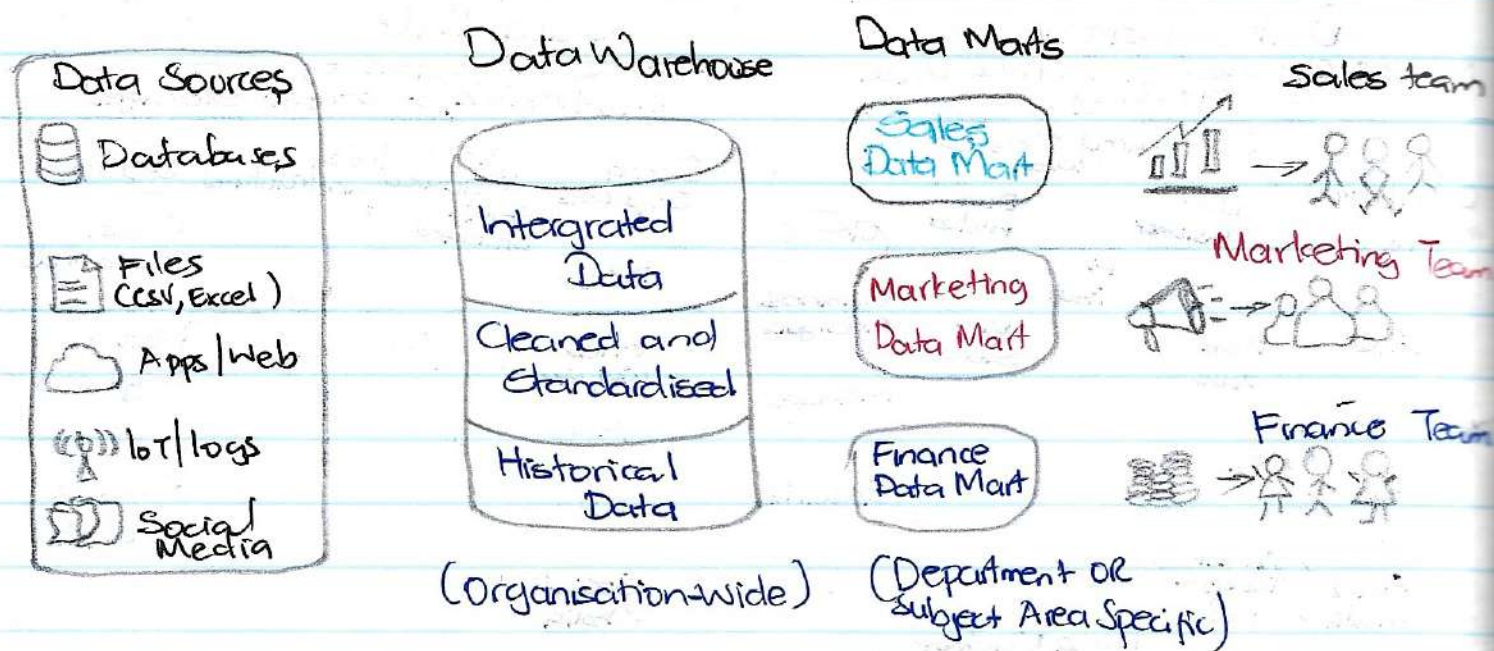
Stores all types of data (Big, flexible, low-cost storage)



Question 4

Data Mart

- i) This is a smaller, focussed version of a Data Warehouse. It contains only the data that is relevant to a specific department or business unit rather than the entire organisation.
- ii) It is related to a Data Warehouse because it is usually built from a Data Warehouse. Data Warehouse holds all the data and the Data Mart pulls out only the relevant portion of a specific team.
- iii) It can be used by: finance department, HR dept. or Sales team.



Question 5

Differences between Structured, Semi-structured and Unstructured Data with examples of each

See table in the next page

Structured Data	Semi-Structured Data	Unstructured Data																
Data is organised, follows a fixed format and is stored in rows and columns (like a table)	Data has some organisation (e.g tags, keys, labels, etc) but does not follow a strict format like table	Data has no predefined format and is not organised. It can be text, images, audio, video e.g...																
Examples: Excel/ database table Customer records, Sales transactions	Example: JSON file or XML, web logs, Emails, API's (data with structure but flexible)	Examples: Images, Docs, Audio, videos, social media content, customer reviews, free text documents																
<table><tr><th>Customer ID</th><th>Name</th><th>Email</th><th>Purchase Amount</th></tr><tr><td>MH101</td><td>Tshidi Sefara</td><td>Tsefara@gmail.com</td><td>55.00</td></tr><tr><td>MH102</td><td>Mollie Tsotetsi</td><td>MT123@ABC.com</td><td>48.00</td></tr><tr><td>MH103</td><td>Karl Moore</td><td>MooreK@200t.com</td><td>57.00</td></tr></table>	Customer ID	Name	Email	Purchase Amount	MH101	Tshidi Sefara	Tsefara@gmail.com	55.00	MH102	Mollie Tsotetsi	MT123@ABC.com	48.00	MH103	Karl Moore	MooreK@200t.com	57.00	<pre>"customerID": "MH101", "name": "Tshidi Sefara", "email": "Tsefara@gmail.com", "purchases": [{ "item": "laptop", "amount": 55.00 }, { "item": "Mouse", "amount": 12.50 }]</pre>	 Images (JPG, PNG)  Text Doc  Audio (MP3, WAV)  Video (MP4, MOV)  Social media (WhatsApp, Tweets)
Customer ID	Name	Email	Purchase Amount															
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MH102	Mollie Tsotetsi	MT123@ABC.com	48.00															
MH103	Karl Moore	MooreK@200t.com	57.00															

Question 6

(i) Data modelling

It is a process of designing a structure for your data. It defines how data is organised, what it looks like and how different pieces of data are related.

(ii) It is important because:

- It improves data quality and consistency
- Reduces duplication
- Makes data easier to understand

d) Supports better decision making

e) Helps with scalability and performance.

ii) Two types of data models

a) Conceptual Data Model

Shows the big picture. It defines the main entities and their relationships using simple diagrams

- It is not concerned with technical details like data types or keys



b) Physical data model

Shows exactly how the data will be stored in the database

- It includes data types, primary/foreign keys, indices and other technical details

Specific to the database system

Customer	Order
Cust ID	Order ID
Name	Cust ID
Email	Order Date
Phone	Total Amount

Question 7

Data Mining

- The process of discovering useful patterns, trends and relationships in large amounts of data.

- It uses statistical, machine learning and data techniques to turn raw data into valuable insights

ii) How it is used in Business

a) To understand customer behaviour

b) Identify new revenue opportunities

c) Manage risks and detect fraud

d) Improve products and services

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iii) Two real-world examples of Data Mining

a) Retail - customer purchase patterns

A supermarket uses data mining to analyse what customers buy together (Association rules)

eg, Customer buys bread, milk and eggs - Supermarket does data mining to find a pattern (If bread + milk then eggs) → then supermarket comes up with special offer (Bread + eggs + milk) then targeted promotion increases sales.

b) Banking and fraud detection

Banks mine millions of transactions daily to detect unusual patterns.

If your card is suddenly used in 3 different countries in one hour, data mining flags it as a fraud immediately.

Question 8

i) DBMS (Database Management System)

- It is a software that allows users and organisations to store, create, organise, manage, retrieve and protect data from a database.

- It acts as a middleman between the user and the database

ii) Four popular DBMS platforms and their suitable uses

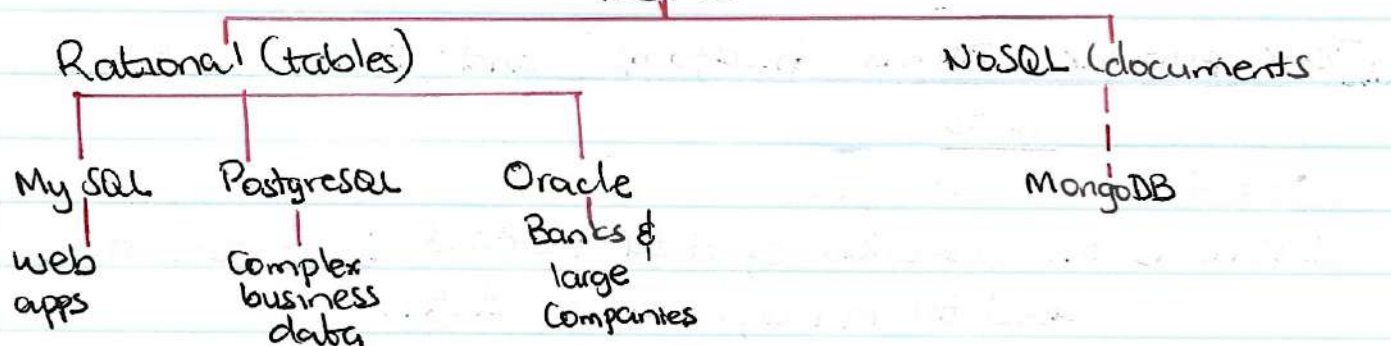
SEE the next page!!

Database is the collection of data, a DBMS is a software used to manage that data.

DBMS Platforms

DBMS	Type	Description	Suitable use
① MySQL	Relational (SQL)	popular, open-source database that stores data in tables and uses SQL	Websites and e-commerce (storing customers, products, orders and payments)
② PostgreSQL	Relational (SQL)	Powerful open-source database with strong standards compliance and advanced data capabilities	Business applications and analytics where complex queries and reliable data management is needed
③ Oracle Database	Relational (SQL)	Enterprise-grade DBMS designed for large, complex and highly critical databases	Banks and large corporations handling millions of transactions where security, reliability and availability are essential
④ MongoDB	NoSQL / Document	Stores data as flexible documents rather than transactional rows and columns	Applications with changing or flexible data, (eg social media, mobile apps or product catalogues)

DBMS



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Question 9

(i) ETL (Extract, Transform, Load)

It is a process used to take data from different sources, clean and prepare it and then load it into a system where it can be analysed.

Step 1 : Extract

Raw data is pulled out from multiple sources
i.e. Databases, Excel spreadsheets, APIs, CSV files, websites, emails, CRM systems.

The data at this step is messy, inconsistent and comes in different formats.

Step 2: Transform

The extracted data is cleaned, standardised and reorganised into a consistent format ready for analysis. i.e. removing duplicates, correcting errors, converting formats (dates, currencies), changing data types, renaming columns, etc.

Step 3: Load

This means putting the transformed data into a destination system where it can be stored and analysed.

- destination can be: data warehouse, database, data lake or an analytics platform.

(ii) ETL is important in data analytics because without ETL analysts would be working with raw messy inconsistent data therefore making their insights unreliable.

ETL helps provide a better data quality

ii) Consistency

iii) One reliable source

Question 10

- i) Common data storage formats used in analytics.
- The different ways data can be saved and structured in a file.
 - Choosing the right format affects how fast data can be read, how much storage it uses and what tools can work with it.

ii) FOUR COMMON FORMATS

Ⓐ CSV = Comma Separated Values

this file stores data in rows and columns with values separated by commas.

Example	Use
Tenant, Property, Month, Rent Joan, Property A, Jan-24, 2400 Charles, Property A, Jan-24, 2300 Mercy, Property A, Jan-24, 2300	• When you need a simple and widely supported format • When working with Excel, PowerBI or small datasets • When sharing data between different systems

Ⓑ JSON = JavaScript Object Notation

→ Stores information in key value pairs using curly brackets and can handle nested or hierarchical data.

→ Flexible and human readable

EXAMPLE	USE
<pre>{ "tenant": "Karabo", "property": "Property A", "rent": 2400, "month": "Jan-24" }</pre>	• When working with APIs and websites • When you need nested information • When data has a complex or changing structure

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(c) Parquet

→ A compressed column based storage format designed for big data analytics.

→ Very fast and efficient.

EXAMPLE

Business / Organisation storing millions of sensor readings efficiently.

USE When:

- Working with large datasets
- You frequently query only certain columns

Parquet / Column-based

Customer → Customer → Customer

Product → Product → Product

Date → Date → Date

Amount → Amount → Amount

(d) Avro - Row-Based Data format

→ Designed for fast data transfer between systems

→ Great for streaming data

- Stores data in rows and uses a schema to describe what the data looks like.

EXAMPLE

- Schema
 - ├ Customer → Text
 - ├ Product → Text
 - ├ Date → Date
 - ├ Amount → Number

- Bank transactions being streamed and processed instantly.

WHEN TO USE:

- When data need to move quickly between systems in real time
- Building data pipelines
- Working with technologies such as Apache Kafka or Hadoop
- You need a clearly defined schema

SECTION B: ARTIFICIAL INTELLIGENCE & ML

Question 11

(i) Artificial Intelligence

- The ability of computer system to perform tasks that normally require human intelligence such as learning, reasoning, problem solving, understanding language and recognising patterns.

(ii) Differences between Narrow AI (Weak AI) and General AI (Strong AI)

<u>Narrow AI (Weak AI)</u>	<u>General AI (Strong AI)</u>
• Designed for a specific task • Limited Capabilities • Cannot easily transfer knowledge to unrelated tasks • Exists today <u>Example: Siri</u> → Siri or Google Assistant → They answer questions brilliantly but cannot drive a car or diagnose an illness	• Designed to handle many different tasks. • Broad, human-like capabilities • Would be able to transfer and apply knowledge across tasks • Not yet achieved <u>Example:</u> → A hypothetical AI → A robot that can simultaneously diagnose patients, write code, drive a car and hold conversation

Question 12

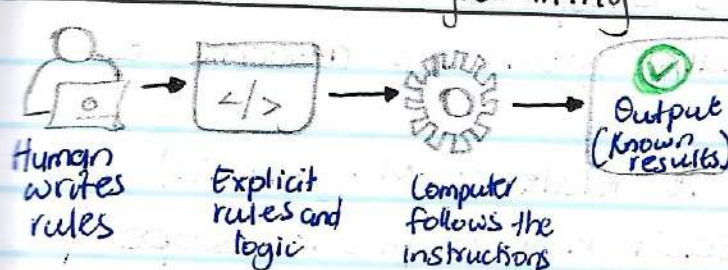
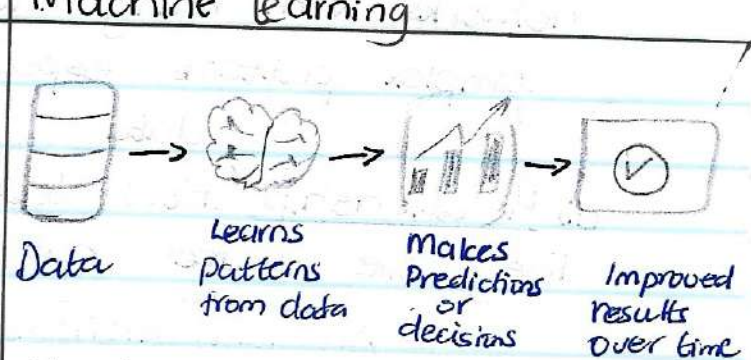
(i) Machine Learning (ML)

- Is a branch of AI that allows computers to learn from data, find patterns and make decisions or prediction without being explicitly programmed for every task.
- Instead of following fixed rules, the machine finds

(13)

patterns in data and learns from experience.

ii) How it differs from Traditional Programming.

Traditional Programming	Machine Learning
 • Uses fixed rules and logic • Work well when the problem is clearly defined • Does not learn from data	 • Handles complex and changing data • Learns from data instead of being explicitly programmed • Finds patterns and improves over time

(iii) Three main types of ML

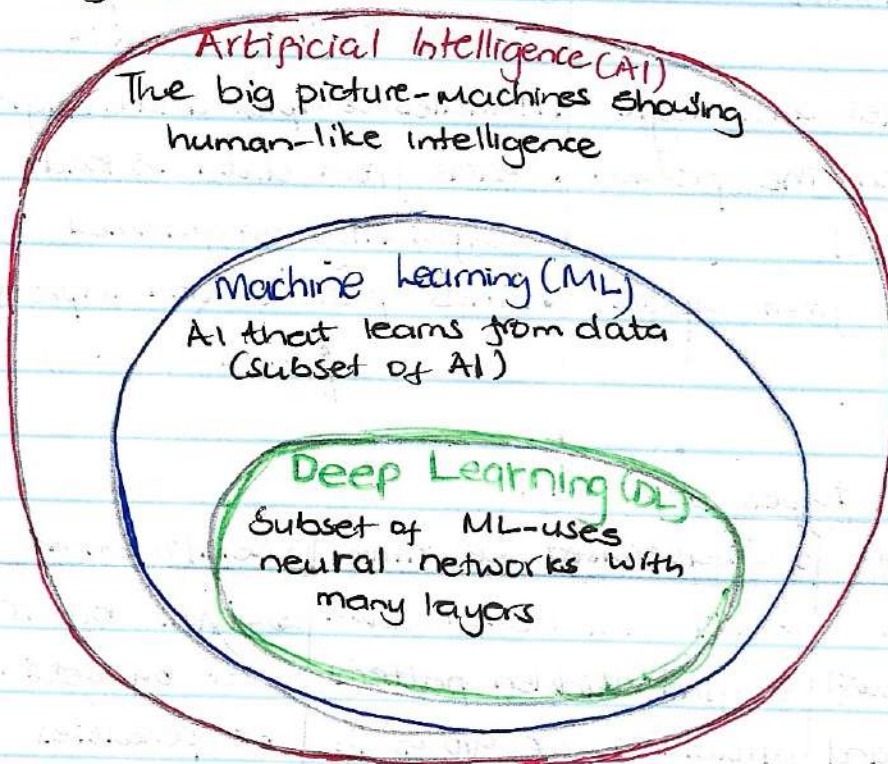
① Supervised Learning	② Unsupervised Learning	③ Reinforcement Learning
⇒ Uses labelled data (input + correct output) to learn a pattern and make predictions on new data Ex. Spam email detection • Used for classification and regression tasks • Requires labelled data	⇒ Uses unlabelled data to find hidden patterns, structures or group in the data. Ex. Customer segmentation • Used for clustering, association and dimensionality reduction • No labelled data needed	⇒ Learns by trial and error by getting rewards or penalties for its actions. Ex. Game playing / Robotics • Used for decision-making and optimisation • Learns by interacting with the environment.

In Short: ML uses data to learn, while traditional programming follows rules. Supervised learns with labels, Unsupervised find patterns and Reinforcement learns through rewards

Q13.ii Deep Learning

→ A specialised form of ML that uses artificial neural networks with many layers (deep networks) to learn complex patterns especially from large amounts of data

iii Relationship (How does it relate to ML and AI)
They are related but each has different scope.



* Deep Learning is AI and Machine Learning but not all AI and ML is DL.

* DL is a special type of ML, and ML is a part of AI.

iii Neural Networks and how they work.

→ Neural Networks are computer systems inspired by human brain. They consist of layers of interconnected nodes (neurons) that process information and learn from data.

b) How do they work? → ① Input data → the network

(15)

receives data and converts it into numbers

② Forward pass $\Rightarrow$ The data moves through the layers. Each node performs a calculation and passes the result forward.

③ Calculate error $\Rightarrow$ The output is compared to the correct answer. The difference is the error.

④ Backpropagation $\Rightarrow$ The error is used to adjust the weights in the network, so it can make better predictions next time.

⑤ Repeat $\Rightarrow$ This process happens many times until the model learns the patterns in the data.

Question 14 (i) Natural Language Processing (NLP)

$\rightarrow$ is a branch of AI that enables computers to understand, interpret, process and generate human language in the form of text and speech.

$\rightarrow$ It combines computer science, linguistics and ML to make sense of the way we communicate.

(ii) Three real world applications of NLP

(a) Virtual Assistants

Siri, Google assistant and Alexa use NLP to understand your voice command and respond naturally.

(b) Text messaging and Translation

NLP powers predictive text, autocorrect and language translation apps by understanding what we mean and the context of our words.

(c) Search Engines $\rightarrow$ NLP helps search engines understand the meaning behind your search, not just the keywords, giving

you more accurate results.

Question 15 (i) Large Language Model (LLM)

→ LLM is a very advanced type of AI model trained on massive amounts of text data (books, websites, articles, code) to understand and generate human language.

(ii) It differs from standard ML (see table below)

Standard ML Model

- Trained for one specific task
- Limited to its training category
- Smaller data set
- Predicts numbers or categories

LLM

- Trained on vast general knowledge
- Can handle many different tasks
- Billions of words of training data
- Generates human-like text and conversation

Standard ML model is like a student who studied one specific subject. An LLM is like a student who read every book in every library in the world and can talk about almost anything.

(iii) Two well known LLMs and their uses

a) ChatGPT by Open AI

→ ChatGPT is one of the most widely used LLMs in the world. Used for writing assistance, answering questions, coding help, summarising documents and creative writing.

b) Claude by Anthropic

→ Claude is also one of the most widely used LLMs. Used for research assistance, document analysis, career coaching, assignment help.

(17)

Question 16

i) Generative AI

This is a type of AI that can create new content, such as text, images, music, videos, code and more, based on patterns it has learned from existing data.

- Unlike traditional AI that just analyses or classifies data, Generative AI actually produces something new and original.

(ii) How does it work at a high level?

- Generative AI is trained on massive datasets of existing content.

- It learns the patterns, styles and structures within that data.

- When it is given a prompt, it uses the learned patterns to generate something new that matches what has been asked.

(iii) 3 tools that use Generative AI

Claude and ChatGPT	DAV-E and Midjourney	GitHub Copilot
→ Generate human like text → Answers questions → Writes essays → Drafts emails → Helps with assignment	→ Generate completely new images from text descriptions	→ Generates code automatically as developers type → Suggests entire functions and solutions

Q 17(i) Training dataset

→ A collection of data used to teach an AI or Machine Learning model to identify patterns and make predictions.

→ AI model needs a training dataset to learn patterns, make predictions and generate outputs.

(ii) Quality of training dataset is important because:

→ an AI model can only learn from the information it is given.

Quality training dataset must be:

- Accurate > correct and verified information.
- Complete > no important gaps or missing values
- Relevant > related to the problem being solved
- Consistent > data follows the same standards

(iii) What could go wrong if the training data is biased or incorrect?a) Biased Hiring AI

An AI trained mostly on male cvs learned to favor male candidates and automatically rejects women's applications.

b) Biased Facial recognition

AI trained mostly on light skinned faces struggled to accurately recognise darker skinned faces leading to wrong identifications.

c) Medical misdiagnosis

An AI trained on medical data from one population group gave incorrect diagnoses when used on patients from different backgrounds because training data did not represent everyone

Q18 (i) Computer Vision

→ This is a field of AI that enables machines to see, interpret and understand visual information from the world (i.e. images, videos, live camera feeds) just like human eyes and brain work together.

→ The goal is to teach computers to extract meaningful information from visual data and make decisions based on what they see.

(ii) How do machines 'see' and interpret images?

A computer sees an image as millions of tiny pixels, each pixel is just a number representing colour and brightness.

→ Computer vision uses Deep Learning and neural networks to analyse all those numbers and find patterns that match ~~to~~ what it has ~~learning~~ learned during training.

~~Image Input - Pixel analysis - Feature Extraction - Pattern~~

- Image Input - camera or photo feeds into the system
- Pixel analysis - image broken into millions of numerical values
- Feature Extraction - Neural network identifies edges, shapes, colours, textures.
- Pattern Matching - compares features against learned patterns
- Output - produces a result ("This person is Mologadi")

(iii) Two real-world use cases of computer vision

Facial recognition on smartphones

iPhone uses computer vision to scan your face, map 30000 invisible dots across your feature and compare them to your stored face data

Medical Imaging in Hospitals

Medical Computer vision analyses X-rays, MRI scans and CT scans to detect tumours, fractures and diseases, sometimes spotting things the human eye misses

SECTION C

Q19 w API (Application Programming Interface)

- API is a set of rules and protocols that allows different software applications to communicate and share data with each other.
- It acts as a messenger or bridge between two systems

(ii) Real world analogy

Electricity meter ⇒ When Eskom/municipality wants to know how much electricity your house used this month.

- They don't come inside the house to check every electrical equipment, the smart meter outside reads your usage and sends that specific information to Eskom/municipality system automatically.
- ∴ The meter is the API → It sits between your house and municipality, sharing only information that is needed.

(iii) Why are APIs important in modern software and data systems

- a) Connection → allows different systems to talk to each other even if built differently
- b) Efficiency → no need to rebuild systems from scratch just connect via API
- c) Real time data → apps can pull live data instantly
- d) Security → users access data through controlled API without touching the actual database directly

Question 20 (i) API key.

⇒ An API key is a unique secret code that identifies who is making a request to an API.

→ It is like a password or VIP access card that proves you are authorised to use a particular API key.

(ii) Purpose of API Key

- Identifies who is accessing the API.
- Controls access (only authorised users can use it)
- Tracks usage → monitors how many requests are being made.
- Prevent abuse → stops unauthorised people from overloading the system.

(iii) How to store an API key and consequences of not protecting API

- Never share it publicly or post it on social media.
- Never put it directly in your code that others can see.
- Store it in a secure environment file called `.env` file.
- Treat it exactly like your bank PIN.

What could happen if exposed publicly?

- Someone can use your API key to access the system pretending to be you.
- You could be charged for their usage.
- Account can be suspended or banned.
- Sensitive data can be stolen or leaked.

Real world example

Q21. REST API (REpresentational State Transfer)

(i) A set of rules and standards for building APIs that are simple, lightweight and work well over the internet.

→ A REST API follows these rules to allow systems to communicate efficiently.

(ii) The four main HTTP methods

- **GET** (read a file) / ~~Create (CRUD)~~

- Retrieves or reads data. It is like opening the filing cabinet and reading a tenant's file without changing anything.

Ex = "Show me all tenants in Property M1101".

- **POST** (Add new file)

→ Creates new data. (like adding a brand new tenant file into the filing cabinet)

Ex = "Add new tenant Agnes to the system".

- **PUT** (Update a file)

→ Updates existing data (like replacing an old tenant file with updated information)

ex = "Update Mologadi's rental amount from R2300 to R2500"

- **DELETE** (Remove a file)

→ Removes data. (like taking out and throwing away a tenant file that's no longer needed.)

ex = "Remove tenant who has vacated the property"

Q22. JSON (JavaScript Object Notation)

→ JSON is a lightweight and human readable data format used to store and exchange data between systems.

- Despite having JavaScript in its name, it works with almost every programming language including Python.

JSON can be taken like a universal language that is used by different systems to talk to each other. Like English being used between people who speak different home languages.

(ii) JSON is commonly used when working with APIs and data exchange for the following reasons:

- Human readable - easy for both humans and machines to read and understand.
- Lightweight - small file size so data transfers quickly.
- Universal - works with almost every programming language.
- Flexible - can represent simple and complex data structures.
- Standard format - most APIs send and receive data in JSON format.

(iii) Example of JSON (Simple)

```
{
  "property": "Mamarikle Holdings",
  "address": "6698 Nkululeko Street",
  "tenant": {
    "id": "MH101",
    "name": "Papiki Mmurwa",
    "room": "M10013",
    "rental_amount": 2000,
    "status": "Active"
  }
}
```


Question 23 Prompt Engineering

→ The skill of crafting clear, specific and well structured instructions to get the possible output from an AI model.

→ The better your prompt the better the AI's response

vii Why is it an important skill?

- Gets better results - clear prompts lead to more accurate, relevant and useful answers.
- Saves time - less back and forth, fewer follow-up questions
- Unlocks more features - helps you get creative, detailed and complex responses.
- Builds your AI confidence - you learn to use the tool effectively and creatively.
- Gives one a competitive edge - a valuable skill in school, work and everyday life

(iii) Two ~~ex~~ examples of a poorly written prompt and an improved version.

✗ **Poor prompt:** "Write me a CV"

✓ **Improved:** "Write a professional CV for Mologadi Shiri, a Data Analyst in training with 16 years of experience in energy and operations at Eskom, holding a BTech in Operations Management and ISO 9001:2015 certification, applying for a remote data analytics internship",

✗ **Poor prompt:** "Analyse my data"

✓ **Improved:** "Analyse this property rental income data for January to December 2026, identify which month had the highest profit, which property performed best and highlight any months where expenses

exceeded income."

"Tell me about climate change"



"Explain the main causes of climate change, and give 3 real-world examples of how it affects people, especially in coastal areas keep it simple and clear."

Question 24. MAP function

- A MAP function applies the same operation or transformation to every single item in a list automatically without you having to write a loop for each item individually.
- It processes each element and returns a new transformed result.

iii) What it does in programming and data processing

Programming

- It applies the same function to every item in a list and returns a new list with the results.

List of monthly rentals in Rand
rentals = [2000, 2200, 2300, 2400, 2500, 2300]

Function to add 10% annual increase

```
def increase_rent (amount):
```

```
    return amount x 1.10
```

MAP applies the increase to all rentals @ once

```
new_rentals = list(map(increase_rent, rentals))
```

```
print(new_rentals)
```

Output: [2200.0, 2530.0, 2530.0, 2640.0, 2530.0]

Data Processing

MAP function is used to:

- Clean and transform large datasets quickly
- Convert data formats across entire columns
- Apply calculations to thousands of rows simultaneously
- Standardise inconsistent data entries.

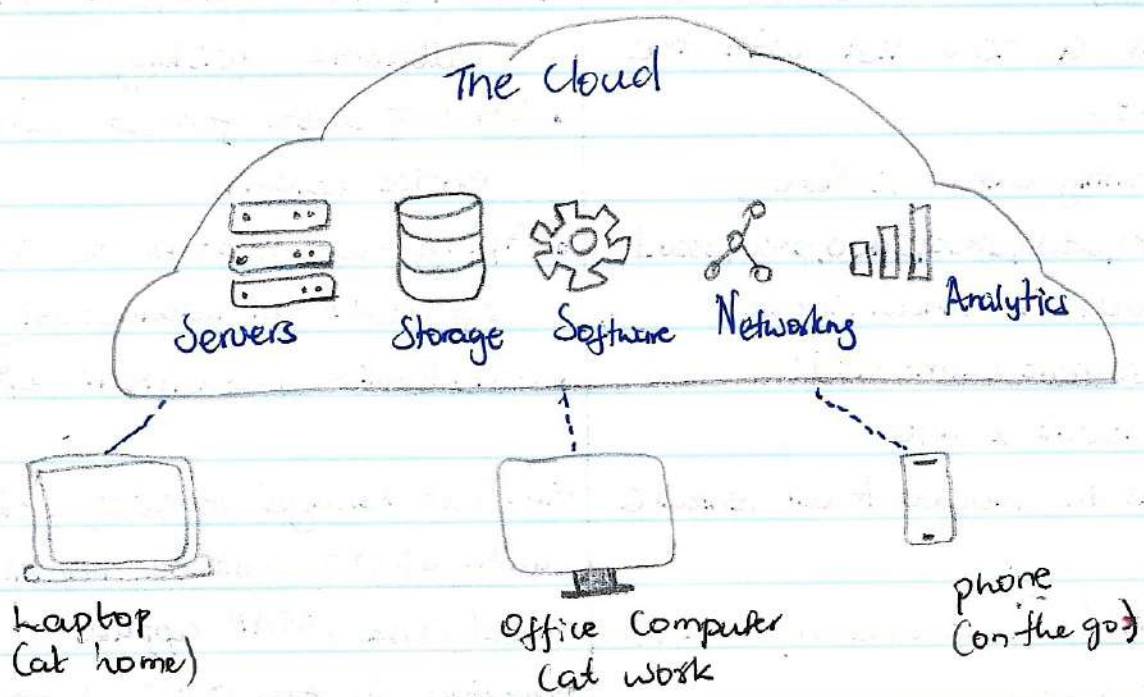
Ex. Data Analyst receives a dataset with 50000 customer names in mixed cases. MAP applies a clean function to all 50000 names simultaneously making them all consistent.

Question 25

i) Cloud Computing - is the delivery of computing resources (such as servers, storage, databases, networking, software and more) over the internet, instead of owning and maintaining physical hardware yourself.

iii) 3 main cloud service models

Iaas (Infrastructure as a Service)	Paas (Platform as a Service)	Saas (Software as a Service)
Provides the basic infrastructure you need in the cloud (virtual servers, storage, networking, etc)	Provides a complete platform for building, running and managing applications (operating system, runtime, databases, development tools)	Provides fully developed software applications over the internet.
Example's Amazon (Virtual server in the cloud)	Example's Microsoft Azure (App Service) Platform to build and run web apps	Example's Google Workspace (Gmail, Docs, Drive)



Question 26 Version Control System

This is a tool that tracks and manages changes made to files and code over time.

- It keeps a complete history of every change so you can go back to any previous version if something goes wrong.

Git - is a popular, open-source version control system. It is designed to be:

- * Fast and reliable
- * Works offline (on your computer)
- * Widely supported (e.g. GitHub, GitLab, Bitbucket)

Reasons why Git is widely used by data analysts and developers?

- * Tracks every change made to code, analysis files or scripts.
- * Allows teams to collaborate without overwriting each other's work.
- * Helps you experiment without breaking your main work.
- * Free and works with GitHub for online storage
- * Useful for data analysts, reporting, dashboards and software development.

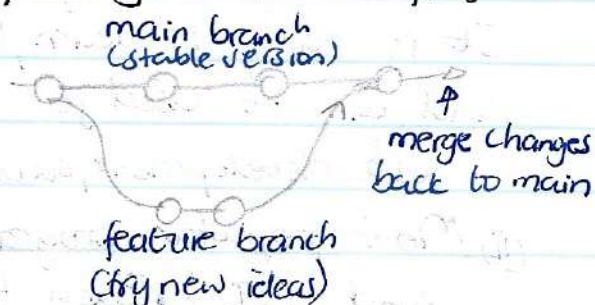
iii

Commit

- a snapshot of your changes at a specific point in time
 - It saves your changes with a message (e.g. "Added data cleaning script")
 - Each commit has a unique ID
 - Commits help you keep a clear history of your work

branch

- separate copy of your project where you can make changes and experiment without affecting the main project



Question 27 w/ Jupiter Notebook

- Is an open-source, web-based interactive environment that allows users to write and execute code, view results, create visualisations and add explanatory text in one document.
- It is most commonly used with Python and is popular among Data Analysts and Data Scientists because it makes data cleaning, analysis, visualisation, experimentation and sharing easy and interactive.

(ii) Programming language commonly used with:

- Jupiter Notebook is mostly used with Python.
- While Python is the most popular, Jupiter also supports other languages such as R, Julia and SQL.

(iii) I.T is popular among Data Analysts and Data Scientists for the following reasons:

- Interactive (Run code and see results immediately)
- Great for data analysis (easily explore, clean and visualise data)
- Shareable (Notebooks can be shared with others via GitHub)
- Widely used in the data community - supported by many libraries and resources.
- Combines code and commentary - add explanations, notes and charts in the same file

Question 28 (i) Python

- It is a high-level, open source programming language.
 - It is easy to learn, has a clean and readable syntax, and can be used for many purposes including data analysis, web development, automation and AI.
- (ii) Most popular programming language for Data Analytics and AI & Easy to learn and use

- Powerful libraries (a wide range of tools for data analysis, visualisation and AI).
- Huge community - millions of users worldwide, many tutorials examples and help available.
- Versatile - used for data analytics, machine learning, AI, automation and more.
- In demand - widely used in industries, making it a valuable skill for your career.

iii) Four Python libraries commonly used in data work:

- Pandas** $\Rightarrow$ used for data manipulation and analysis.
 - $\Rightarrow$ Helps you work with structured data (i.e. tables)
- NumPy** $\Rightarrow$ used for numerical calculations.
 - $\Rightarrow$ Provides support for large, multi-dimensional arrays
- Matplotlib** $\Rightarrow$ used for data visualisation
 - $\Rightarrow$ Helps you create charts and graphs
- Scikit-learn** $\Rightarrow$ used for machine learning
 - $\Rightarrow$ Provides simple tools to build and evaluate AI models.

SECTION D DATA ETHICS, GOVERNANCE & CAREERS

Question 29. i) Data Privacy

- Is about the proper collection, use, storage, sharing and protection of personal information.
- It gives individuals control over their own data and ensures it is handled responsibly and securely.



(ii) Why data privacy is important

- Protect individuals from identity theft and fraud.
- Prevents personal information being sold without consent.
- Build trust between organisations and their customers
- Gives people control over their own digital footprint.
- Better business practices that lead to responsible and ethical use of data.

(iii) a) GDPR (General Data Protection Regulation)

- Data protection law introduced by the European Union (EU) in May 2018.
- It sets rules for how organisations collect, use, store and protect the personal data of individuals.

(b) GDPR applies to:

- All 27 European Union (EU) member states
- The European Economic Area (EEA) countries (EU + Iceland, Liechtenstein and Norway)
- It also applies to organisations outside the EU if they offer goods or services to people in the EU, or monitor their behaviour

Question 30 is Data Ethics

- A set of moral principles and guidelines that govern how data is collected, stored, analysed and used responsibly.
- It ensures that data is used fairly, transparently and without causing harm to individuals or society.

(i) Examples of unethical use of data or AI

a) Biased AI in hiring

- A company uses an AI tool to screen job applications. The tool is trained on past data that favours male

(21)

candidates, so it consistently ranks men higher than equally qualified women.

PROBLEMATIC BECAUSE - It is unfair and discriminating.

- It reinforces existing bias
- It can lead to unequal opportunities and harm people's careers.

(b) Selling medical data without consent

Health organisations have been found selling patient medical records to pharmaceutical and insurance companies without patients' knowledge or consent, allowing the insurance companies to deny coverage based on pre-existing conditions.

PROBLEMATIC BECAUSE - Violates patient confidentiality and trust.

- People can be discriminated against based on their health history
- Patients never agree for their data to be used this way.

(c) Misuse of personal data for targeted advertising

A social media platform collects users' personal data (eg. browsing history, location, interests) and uses it to show highly targeted ads without clear consent. This data may also be shared with third parties.

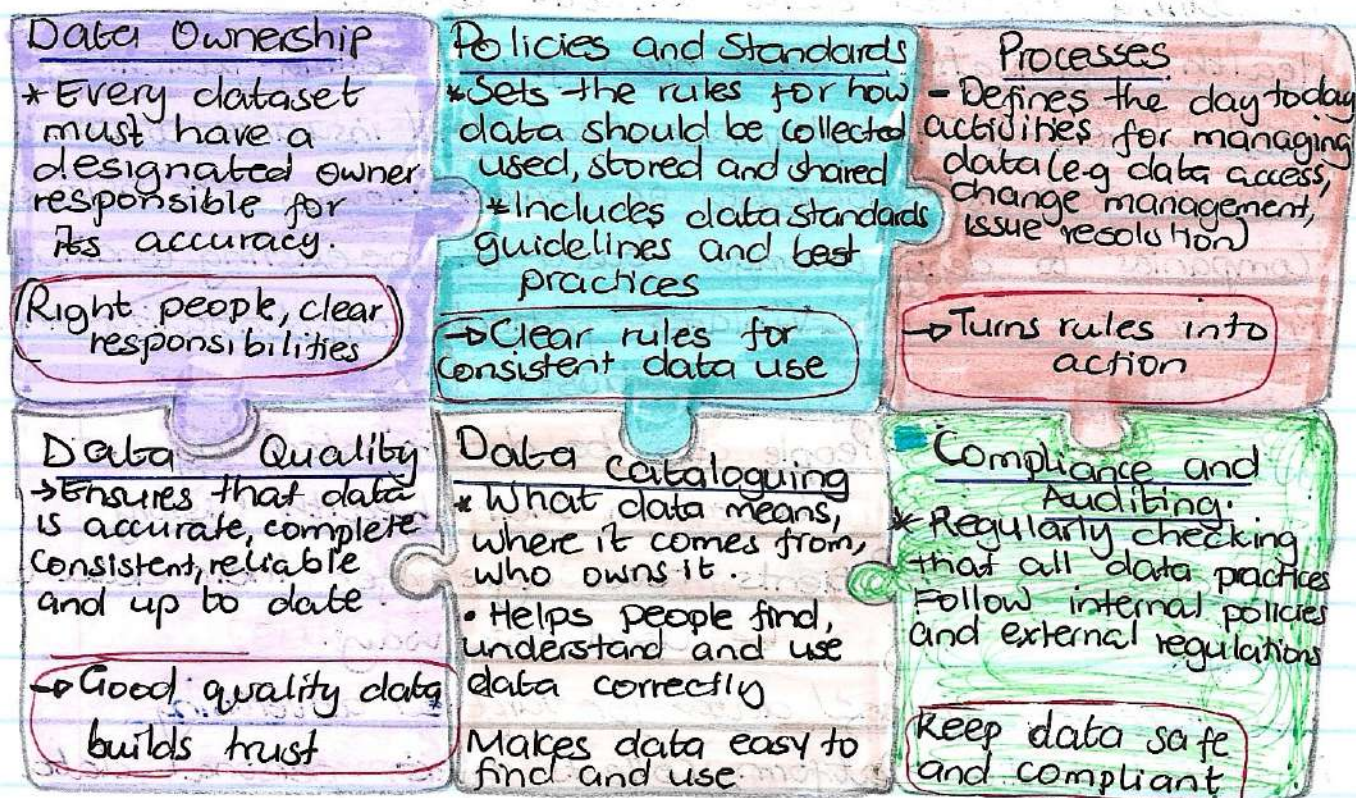
PROBLEMATIC BECAUSE - People may not know how their data is being used.

- It can feel invasive and manipulative behaviour
- It reduces people's control over their own information
- It can damage trust

Question 31. ii) Data Governance

→ is a framework of policies, processes, roles, standards and controls used to ensure that an organisation's data is accurate, secure, accessible, consistent, compliant and managed effectively.

iii) Key components of a good data governance.



Question 32. ii) Bias in AI

→ This refers to systematic and unfair discrimination in the predictions, decisions or recommendations made by an AI system.

→ Occurs when an AI model produces results that consistently favour or disadvantage certain groups of people, usually due to biased data, design or use.

iii) A biased AI model is biased when it learns patterns from data that reflect historical inequalities

or unfair assumptions and then produces outputs that are not fair or equal across different groups (e.g. by gender, race, age or socioeconomic background)

iii. A real-world example of AI bias and potential consequences.

American courts used COMPAS to predict how likely a criminal offender was to reoffend and help judges decide on sentencing and parole.

What went wrong

Why it happened

An investigation found that COMPAS AI was twice as likely to incorrectly flag Black defendants as future criminals compared to white defendants even when their crime history was similar.

The AI was trained on historical criminal justice data which already reflected decades of racial bias in policing and sentencing.

White defendants who later reoffended were more often rated low risk.

The AI learned and amplified those existing biases.

Black defendants who did not reoffend were more often rated high risk.

POTENTIAL CONSEQUENCES

- Injustice - innocent people punished more harshly based on race and gender
- Legal consequences - organisations face lawsuits and fines.
- Loss of trust - public loses confidence in AI system and the organisations using them.

Question 33

1) Key differences between a Data Analyst, a Data Scientist, a Data Engineer and an AI Engineer?

	Data Analyst (Turns data into insights)	Data Scientist Finds patterns and Predicts the future	Data Engineer Builds and manages the data foundation	AI Engineer Builds and deploys AI solution
Role / focus	Analyse and interpret existing data to help business make decisions	Use data, statistics & ML to find patterns, make predictions & solve complex problems	Design, build and maintain the system and pipelines that collect, store and provide access to data	Build, deploy and maintain AI-powered applications
Key responsibilities	- Clean and explore data, Create reports & dashboards, Identify trends and insights - Support business decisions	- Explore & analyse data - Build & evaluate models - Run experiments - Communicate findings - Solve complex problems	- Build data pipelines - Manage databases & cloud architecture - Ensure data is reliable, secure and scalable	- Develop and fine-tune AI models - Integrate AI into apps - Work with LLMs and generative AI - Monitor model performance
Key Skills	- Data analysts - SQL - Data visualisation - Business understanding - Attention to details	- Statistics, ML, Python - R, Data modelling - Problem solving - Communication	Data architecture, ETL/ELT, Databases, Cloud computing, Scripting (Python/Scala), System design	- ML & Deep Learning - Python, Generative AI - LLMs, Model deployment - API development
Common Tools	Excel, SQL, Power BI, Tableau, Google sheets	Python, R, Scikit-learn, TensorFlow, Jupyter Notebook	AWS, Databricks, SQL, Airflow, Python, Snowflake	Python, Hugging Face, Langchain, MLOps, OpenAI, Docker
Example Tasks	- Create sales dashboard - Analyse customer trends - Prepare report for management	- Build a model to predict customer churn - Analyse experiment results - Presents insights to stakeholders	- Build an automated data pipeline - Set up a data warehouse - Ensure data is available and reliable	- Fine-tune a LLM - Build a chatbot or AI application - Deploy the model to production - Monitor & improve performance
Main Outcome	Clear insights and reports to support decision making	Predictive models and advanced insights	Reliable and Scalable data system	Real-world AI solutions that solve problems and create new possibilities

Question 34 (i) Business Intelligence (BI)

→ The process of collecting, analysing, visualising and sharing data to help organisations make informed, data-driven decisions.

→ It uses data, technology and tools to turn raw data into meaningful information, showing what is happening, why it is happening and what might happen next.

(ii) Two popular BI tools

a) Microsoft Power BI

→ A powerful BI tool from Microsoft that helps turn data into interactive reports and dashboards.

How it helps businesses

- Visualises complex data in simple charts and graphs.
- Updates dashboards in real time as new data comes in.
- Allows non technical managers to explore data.
- Connects to many data sources (Excel, SQL, cloud, etc.)

b) Tableau

A leading BI tool known for its powerful data visualisation capabilities.

How it helps the business

- Creates stunning visual stories from complex datasets.
- Drag and drop interface, no coding needed.
- Helps users explore data and find patterns quickly.
- Connects to a wide range of data sources.
- Makes it easy to share insights and tell data-driven stories.

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SECTION E PERSONAL REFLECTION

Q35. The concept I found most interesting and surprising was the STAR SCHEMA and how it relates to real world data organisation. I was fascinated to discover that my own property portfolio tracker at Mamarihle Holdings could be restructured into a proper Star Schema with central Fact table containing payment amounts and surrounding Dims Tables for tenants, properties, dates and payment types.

What surprised me most was realising that this powerful data modelling concept was not only used by large corporations like Amazon and Takealot but was directly applicable to my everyday property management work.

This made the concept feel real and immediately relevant rather than just theoretical.

Q36

The concept I found most difficult to understand was Deep Learning and Neural Networks. The idea of machines learning through multiple hidden layers of interconnected nodes, mimicking the human brain was challenging to visualise at first. To help me understand better I used the BrightLearn study guide, IBM SkillsBuild Data Fundamentals course where I scored 100% and additional research from Google and IBM documentation. Breaking it down into the three layers, i.e. Input, hidden and Output and drawing the neural network diagram helped me grasp the concept more clearly.

Q37 - I see Data and AI playing a central and transformative role in my future career. My goal is to help organisations solve complex operational problems by providing meaningful data driven insights that assist leadership in making better informed decisions

I am specifically targeting energy sector companies like Eskom, Sasol and globally recognised giants like Saudi Aramco where my 16 years of operational experience at Eskom gives me a unique advantage. I do not just understand the data but also understand the environment the data comes from.

I believe Data Analytics will allow me to prevent unnecessary operational delays, reduce expenses and improve efficiency in ways that directly impact an organisation's bottom line. What excites me most is the idea that my insights, numbers on a dashboard could prevent a plant shutdown, save millions in maintenance costs or help an organisation make a strategic decision that changes its trajectory.

With the skills I am building through Brightlearn, IBM skills build and myself directed learning journey, I am positioning myself to contribute meaningfully to the global data economy while gaining the international experience and recognition that comes with it.

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